

Susan (Shuxiao) Zhang, PhD

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🎓 EDUCATION

Doctor of Philosophy, Plant Biology September, 2025
University of California, Davis

Doctor of Philosophy, Biomedical Science December, 2014
University of California, San Diego

Bachelor of Science, Plant Biology June, 2008
University of California, Davis

🧠 PROFESSIONAL INTEREST

Research in abiotic stress response, especially the interaction between different stress responses. Using anatomy-based approaches to study plant development and cell morphology. Development of Course-based Undergraduate Research Experience teaching modules to address classroom equity, engagement, and Plant Awareness Disparity in undergraduate education.

🔬 RESEARCH EXPERIENCE

Post-doctoral fellow, SLAC National Acceleratory Laboratory, Stanford University, Stanford. (Jan. 2026 – Jan. 2028) Develop pipeline and construct cell-level model of plant heat stress response using AI. Conduct research on regulation of cell wall deposition and metal content in nutshell.

Graduate researcher, Department of Plant Sciences, University of California, Davis. (Sep. 2019- Sep. 2025)

Lead research team to conduct transcriptome-wide research on the mechanism behind salinity and drought stress tolerance in pistachio rootstock and the mechanism behind pericarp dehiscence in pistachio fruits. Collaboration in establishing the cause behind internal kernel discoloration in pistachio. Developed and maintained undergraduate research teams, leading undergraduate training and research with weekly team meetings. Develop undergraduate curriculum with interactive and inclusive research-experience based modules and provide classroom instruction.

Post-doctoral researcher, Department of Biochemistry and Molecular Biology, University of Southern California.

(August 2015 – Aug. 2018)

Conducted research on epigenetic changes and regulation in sickle cell anemia, specifically on *PGF* signalling pathway. Developed training modules for master students and mentored completion of master's thesis. Assisted in training technician and processing samples for a large multi-centered clinical trial.

Graduate researcher, Department of Medicine, University of California, San Diego.
(September 2008-December 2014)

Completion of dissertation on functions of novel gene *Nmf9* which causes observable locomotor deficits. Completion of research proposal and associated research for *Nmf9* project. Application and award of Genetics Training Grant.

TEACHING

Sacramento City College

CURRENT: 2025 Plant biology (BIOL412): Taught lecture and laboratory of introductory level plant biology course. Involved in writing of lecture and lab material, developed graded course assessments. Developed and adapted Course-based Undergraduate Research Experience for the community college demography. Adjunct assistant professor.

UC Davis

2022-2025 Troubleshooting stress response through plant anatomy (FRS003): Initiated and developed curriculum for Course-based Undergraduate Research Experience. Co-instructor / lab instructor. Course rating: 4.7/5.

2022-2025 Plants and society (PLS12): Guest lecturer.

2025 Metabolic processes of cultivated plants laboratory (PLS100AL): Guest lecturer / lab instructor.

2025 Environmental interactions of cultivated plants (PLS100C): Guest lecturer.

2018-2021 Plant anatomy (PLB105): Teaching assistant / lab instructor. Instructor evaluation: 3.9 / 5.

UC San Diego

2010 Genetics (BICD100): Teaching assistant.

MENTORSHIP

(Underline = thesis / practicum supervision, * = students with minority group status, *italics* = students who went on to pursue PhD)

Master students

University of Southern California: *Ke Peng, *Xinyan Liang, *Kyueon Park

Undergraduates

UC Davis: *Annamarie Basco, Kaleigh Bedell, Oliver Betz, *Komal Cheema, Alisa Chernikova, *Fiona Dale-Huang, Shaina Eagle, *Rachel Jang, *Rolando Lopez, Corinne Matthews, *Karen Nguyen, Hudson Sheats, *Riya Shergil, *Phuong Tran

UC San Diego: *Camille Monestime

Lab technician

University of Southern California: *Emiliano Huesca, *Trina McLeod

HONORS & AWARDS

2023, campus-wide Chancellor's Award for Excellence in Mentoring Undergraduate Research, UC Davis
2008-current, Phi Beta Kappa Honor Society

GRANTS

2025-2027: USDA NIFA Post-doctoral Fellowship "Conservation NAC gene function in commercially popular nut crops", \$221,000.
2023-2025: USDA NIFA Pre-doctoral Fellowship "Combinatorial effects of salinity and drought stress on root apoplastic barrier development in *Pistacia vera*", \$120,000.
2021-2025: Plant Sciences Departmental Graduate Student Research Award "Secondary cell wall modifications in response to salinity stress in pistachio rootstock", \$80,000.
2024, Katherine Esau Summer Graduate Fellowship
2022-2023, Knowles Agronomy Scholarship, \$500
2022-2023: Henry A. Jastro Research Scholarship Award "The role of programmed cell death and cell wall modification in pericarp ripening events", \$2,815.
2020-2021: Henry A. Jastro Research Scholarship Award "Assessment of xylem development in pistachio rootstock for improved salinity tolerance", \$2,222.
2009-2021 Genetics Training Program: Institutional Pre-doctoral NRSA T32 2009-2012

PUBLICATIONS (underlined names denote students mentored)

1. **Zhang, S.**, McCurry, H., Eagle, S., Jang, R.J., Tran, P., Sheats, H., Lopez, R., Drakakaki, G. Stress-specific plasticity promotes tolerance under combinatorial salt and drought stress in pistachio rootstock. [manuscript in preparation]
2. Wang, M., Santos, M., Eagle, S., Chernikova, A., **Zhang, S.**, Tran, P., Marino, G., Wheeler-Dykes, B., Wilkop, T., Jernstedt, J., Niederholzer, F., Ferguson, L., Drakakaki, G. Identification of key molecular and cellular events of table olive (*Olea europaea* L.'Manzanillo') fruit abscission zone formation during maturation and upon ethephon treatment. [manuscript in preparation]
3. **Zhang, S.**, Wang, M., Eagle, S., Chernikova, A., Bedell, K.M., Tran, P., Lee, C., Adaskaveg, J.A., Wei, Y., Lopez, R., Basco, A., Gordon, P., Blanco-Ulate, B., Monroe, G., Drakakaki, G. Cell layer-specific modification of cell wall is associated with exomesocarp split in pistachio (*Pistacia vera* L.) Nov. 2025. [manuscript in press; doi: <https://doi.org/10.1093/jxb/eraf519>]
4. **Zhang, S.**, Wang, M., Chernikova, A., Eagle, S., Bedell, K.M., Nguyen, K., Blanco-Ulate, B., Jernstedt, J., Drakakaki, G. Difference in Kernel Shape and Endocarp Anatomy Promote Dehiscence in Pistachio Endocarp. *Journal of the American Society for Horticultural Science*. Sep., 2023. [cover article]
5. **Zhang, S.**, Quartararo, A., Betz, O.K., Madahhosseini, S., Heringer, A.S., Le, T., Shao, Y., Caruso, T., Ferguson, L., Jernstedt, J., Wilkop, T., Drakakaki, G. Root vacuolar sequestration and suberization are prominent responses of *Pistacia* spp. rootstocks during salinity stress. *Plant Direct*. Feb., 2021.
6. Shao, Y., Cheng, Y., Pang, H., Chang, M., He, F., Wang, M., Davis, D.J., **Zhang, S.**, Betz, O., Fleck, C., Dai, T., Madahhosseini, S., Wilkop, T., Jernstedt, J., Drakakaki, G. Investigation of Salt Tolerance Mechanisms Across a Root Developmental Gradient in Almond Rootstocks. *Front. Plant Sci*. Jan., 2021.

7. Kalra, VK., **Zhang, S.**, Malik, P., Tahara, SM. Placenta growth factor mediated gene regulation in sickle cell disease. *Blood Rev.* Jan., 2018.
8. **Zhang, S.**, Ross, K.D., Seidner, G.A., Gorman, M.R., Poon, T.H., Wang, X., Keithley, E., Lee, P.N., Martindale, M.Q., Joiner, W.J., Hamilton, B.A. *Nmf9* encodes a highly conserved protein important to neurological function in mice and flies. *PLOS Genet.* Jul., 2015.
9. Berry, A.M., Mendoza-Herrera, A., Guo, Y., Hayashi, J., Persson, T., Barabote, R., **Zhang, S.**, Pawlowski, K. New perspectives on nodule nitrogen assimilation in actinorhizal symbioses. *Funct Plant Biol.* Sep., 2011

PRESENTATIONS (underlined names denote students mentored)

1. **Zhang S.**, McCurry, H., Eagle, S., Jang, R.M., Tran, P., Sheats, H., Lopez, R., Drakakaki, G. Cell-type specific regulation of apoplastic barrier deposition is associated with improved salinity and drought tolerance in pistachio rootstock. 2025 Plant Cell Biology International Conference. Contributed talk by Zhang.
2. **Zhang, S.**, Drakakaki, G., Gordon, P. How do pistachio shells split? *Grow the Valley Podcast.* Jul. 2025. Guest speaker on podcast.
3. **Zhang S.**, Eagle, S., McCurry, H., Jang, R.M., Tran, P., Sheats, H., Lopez, R., Drakakaki, G. Plasticity promotes tolerance under combinatorial salt and drought stress in pistachio rootstock. 2025 UC Davis Plant Science Symposium. Contributed poster & TikTok by Zhang.
4. **Zhang, S.**, Wang, M., Chernikova, A., Eagle, S., Jernstedt, J., Drakakaki, G. Novel mechanisms for pericarp splitting without dehiscence zone in the pistachio fruit. 2024 Conference for the American Society of Plant Biologists. Poster.
5. Eagle, S., **Zhang, S.**, McCurry, H., Tran, P., Jang, R., Drakakaki, G. Physiological Effects of Combined Drought and Salinity Stress on Four Commercially Available Pistachio Rootstocks. 2024 UC Davis Plant Science Symposium, UC Davis Undergraduate Research Conference. Poster.
6. Jang, R., **Zhang, S.**, McCurry, H., Eagle, S., Tran, P., Drakakaki, G. Effects of Combined Drought and Salinity Stress Responses on Suberin in Pistachio Root. 2024 UC Davis Plant Science Symposium, UC Davis Undergraduate Research Conference. Poster.
7. Tran, P., **Zhang, S.**, Lopez, R., Drakakaki, G. Analysis of Pistachio Fruit and Kernel Width in Relations to Endocarp Dehiscence. 2024 UC Davis Plant Science Symposium, UC Davis Undergraduate Research Conference. Poster.
8. Basco, A., **Zhang, S.**, Eagle, S., McCurry, H., Wang, M., Tran, P., Chernikova, A., Drakakaki, G. Tissue-specific changes in the parenchyma cells of pistachio hull leads to its breakdown during maturation. 2024 UC Davis Undergraduate Research Conference. Poster.
9. **Zhang, S.** Anatomy based approach to understanding stress response and development in *Pistacia spp.* 2023 UC Davis Plant Science Department Seminar Series. Contributed talk by Zhang.
10. **Zhang, S.**, Chernikova, A., Eagle, S., Jernstedt, J., Wang, M., Drakakaki, G. Cellular and morphological traits associated with shell split in pistachios (*Pistacia vera*). 2023 VIII International Symposium on Almonds and Pistachios. Contributed talk and poster by Zhang.

11. Chernikova, A., Zhang, S., Bedell, KM, Nguyen, K., Jernstedt, J., Drakakaki, G. Analysis of morphological changes in pistachio cultivars and their relation to endocarp split. 2023 UC Davis Undergraduate Research Conference. Poster.
12. Eagle, S., Zhang, S., Chernikova, A., Wang, M., Jernstedt., Drakakaki, G. Effects of cuticle thickness and parenchyma cell size on *Pistacia vera* hull dehiscence. 2023 UC Davis Undergraduate Research Conference. Poster.
13. **Zhang, S.** Mechanisms of hull and shell split in pistachios. 2022 UC Davis Agriculture and Natural Resources Pistachio Workgroup Meeting. Contributed talk by Zhang.
14. **Zhang, S.** Pericarp maturation and dehiscence in pistachio. 2022 Plant Biology Graduate Group Tuesday Seminar Series. Contributed talk by Zhang.
15. **Zhang, S.,** Quartararo, A., Shao, Y., Cheng, Y., Betz, OK., Madahhosseini, S., Pang, H., He, F., Chang, MC., Wang, M., Brown, PJ., Wilkop T., Jernstedt., Ferguson, L., Drakakaki., G. Investigation of Salinity Response Mechanisms across a Root Developmental Gradient in Almond and Pistachio Rootstocks. 2021 American Society for Horticultural Science Conference. Virtual Conference. Contributed talk by Zhang.
16. Betz, OK., Zhang, S., Abdoloatof, S., Quartararo, A., Hosseini, S.M., Jernstedt, J., Brown, PJ., Drakakaki, G. Characterization of root plasticity in pistachio rootstocks for better nutrient uptake and stress response. 2019 UC Davis Undergraduate Research Conference. Poster.
17. **Zhang, S.,** Poon, T.P., Gorman, M.R., Wang, X., Keithley, E. M., Hamilton, B. A. Highly conserved *Nmf9* gene is essential for vestibular development and functions of limbic, circadian systems in adult mice. 2013 Society for Neuroscience Conference. Poster.
18. **Zhang S.** Understanding locomotor deficits by analysis of *Nmf9* gene. 2013 Cellular and Molecular Medicine Symposium. Contributed talk by Zhang.
19. **Zhang, S.,** Poon, T.P., Gorman, M.R., Wang, X., Keithley, E. M., Hamilton, B. A. Highly conserved gene *Nmf9* impairs vestibular, limbic, and circadian systems. 2012 Society for Neuroscience Conference. Poster.
20. **Zhang S.** The role and function of *Nmf9*, mystery gene. 2011 Neurodevelopment Interest Group Symposium. Contributed talk by Zhang.
21. **Zhang, S.,** Poon, T.P., Gorman, M.R., Wang, X., Keithley, E. M., Hamilton, B. A. *Nmf9* causes nonclassical vestibular dysfunction. 2010 Society for Neuroscience Conference. Poster.

PROFESSIONAL DEVELOPMENT

Foundations in Teaching Microcredential Certificate

May, 2023

Center for Educational Effectiveness, University of California, Davis

Training in lesson planning, active learning, learning assessment, reflective teaching practices, and universal inclusive classroom design.

ACCELERATE Asynchronously Certificate

April, 2021

Center for Educational Effectiveness, University of California, Davis

Training in online and hybrid instruction, inclusive and equitable e-learning environment.

Clinical Research Design & Management Certificate

December, 2016

Regulatory Science Program, University of Southern California, Los Angeles

Training in clinical research trial design, clinical trial management and analysis, and ethics of biomedical research.

MEMBERSHIP

2023-current, American Society for Plant Biologists
2022-2023, International Society for Horticultural Science
2021, American Society for Horticultural Science
2010, 2012- 2013, Member of Society for Neuroscience

SERVICES & OUTREACH

2025-current, Early Career Representative, ASPB Equity, Diversity and Inclusion Committee member
2022-2023, committee member, Plant Biology Graduate Student Association Seminar Committee
2017-2018, Volunteer, Los Angeles March for Science
2010, Volunteer, San Diego Festival of Science and Engineering
2009, Volunteer, San Diego County Animal Services, San Diego Animal Shelter

REFERENCES

Georgia Drakakaki (professor, thesis adviser)
gdrakakaki@ucdavis.edu
Barbara Blanco-Ulate (associate professor)
bblanco@ucdavis.edu
Andrew Groover (professor)
agroover@fs.fed.us